

Internship Project — Week 2

Project Title: Employee Attrition Prediction using Machine Learning

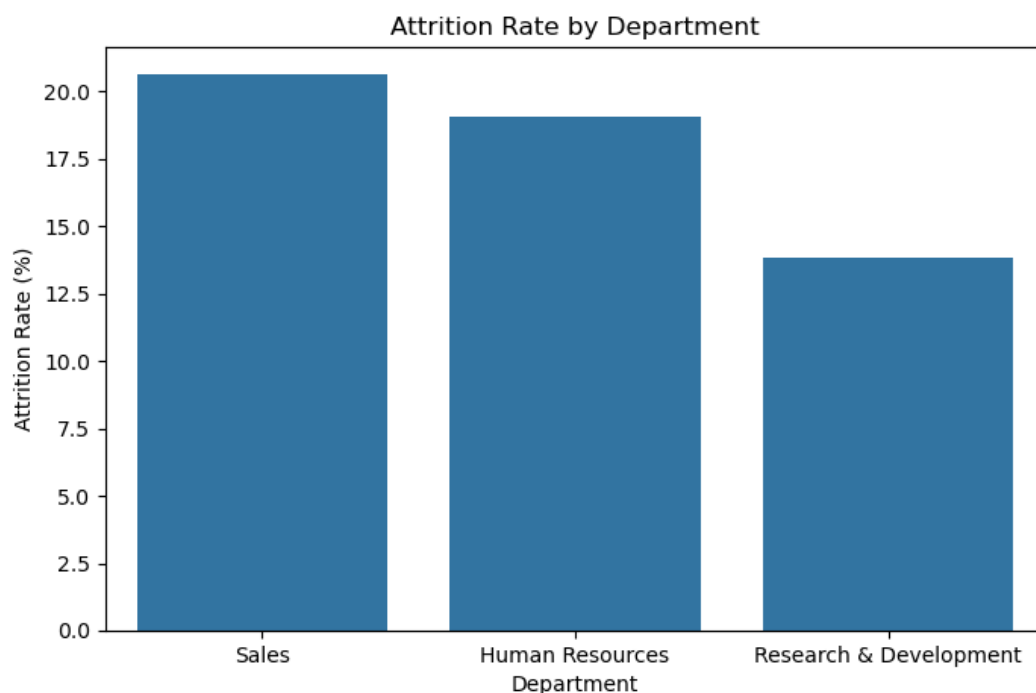
Submitted by: Indrayani Prashant Mude

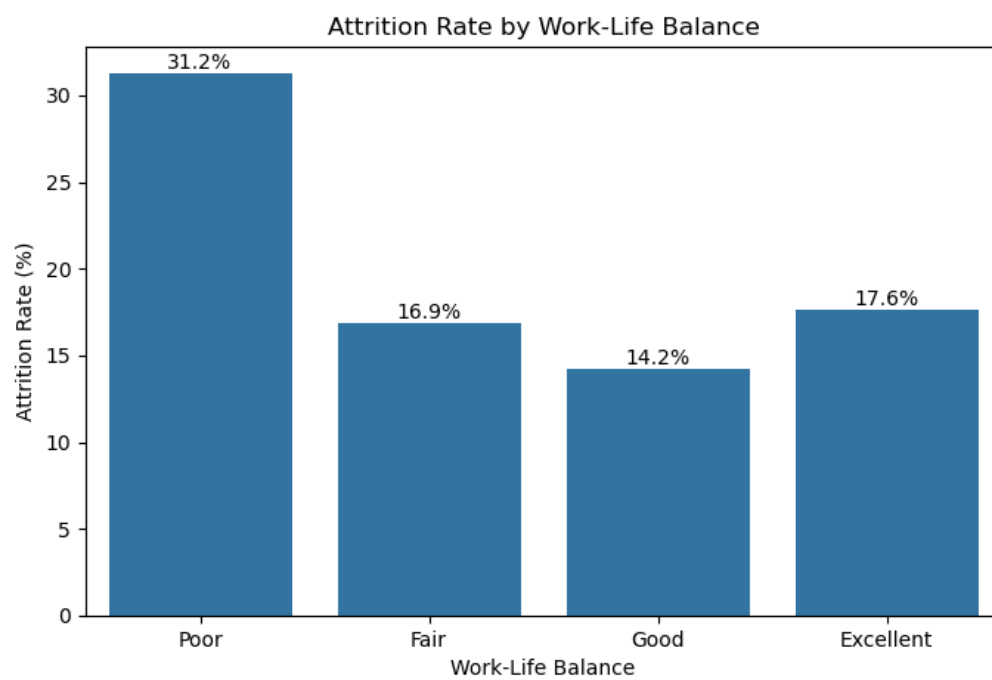
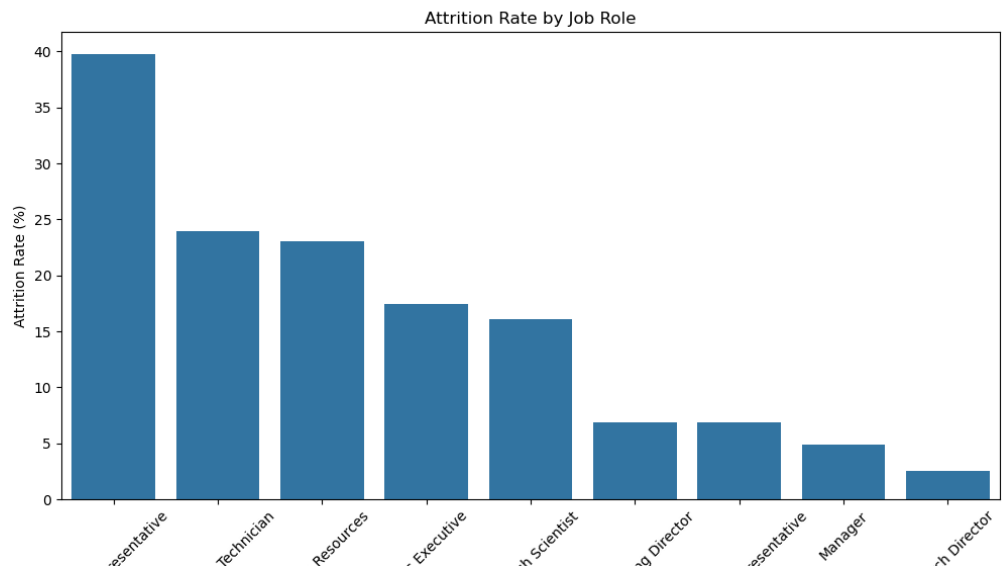
St. Vincent Pallotti College of Engineering and Technology, Nagpur

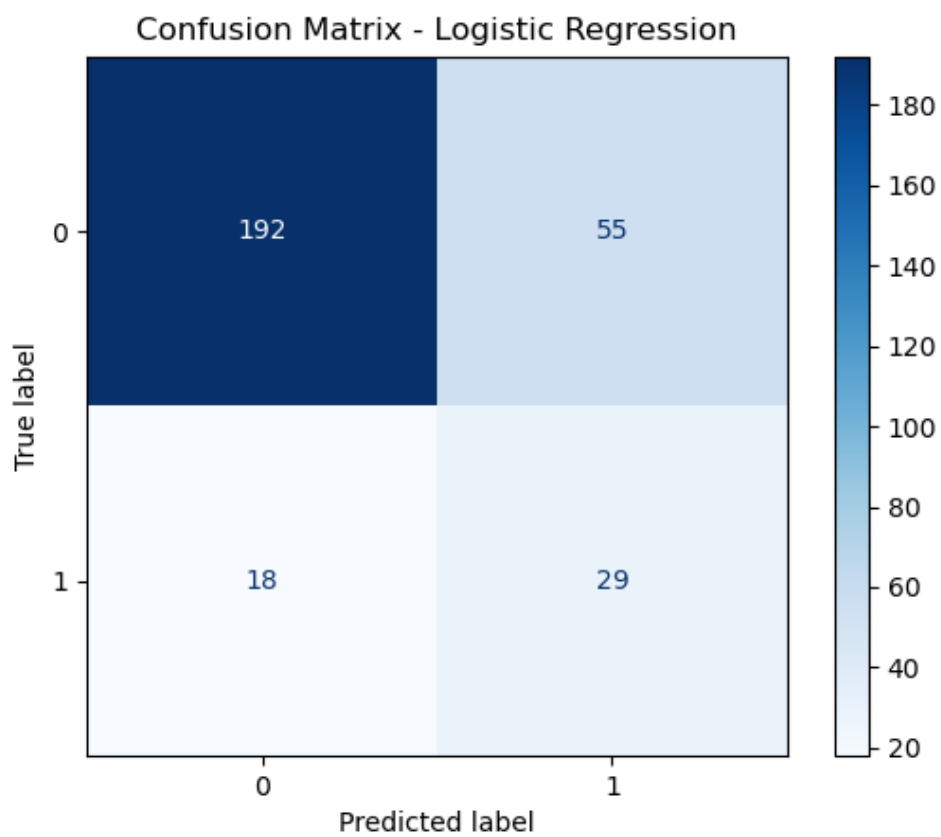
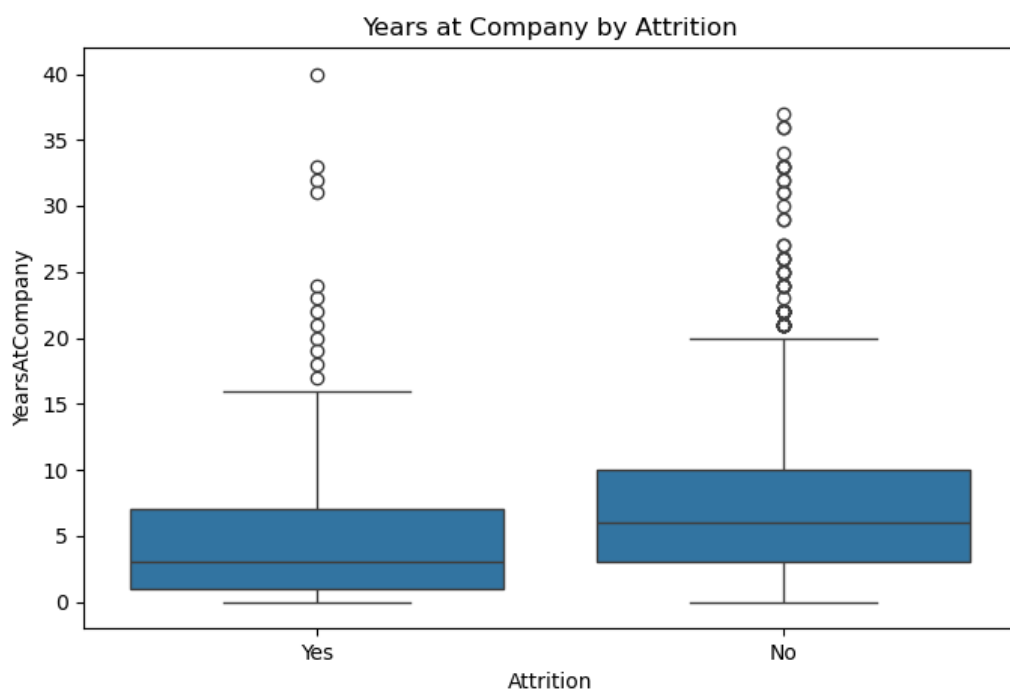
Problem Statement: Every company loses employees — but losing the wrong employees at the wrong time costs the business heavily in hiring, training, and lost productivity. HR departments at large companies spend crores every year trying to figure out who is likely to leave and why — before it happens. This is called Employee Attrition Prediction, and it is one of the most widely used real-world applications of data science in the corporate world today.

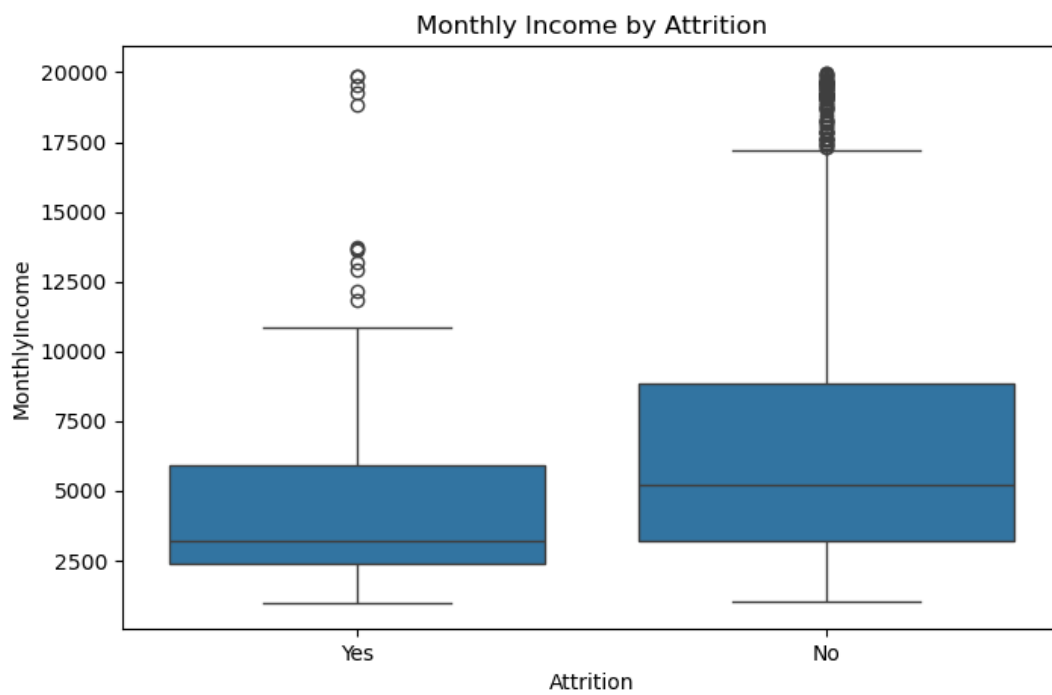
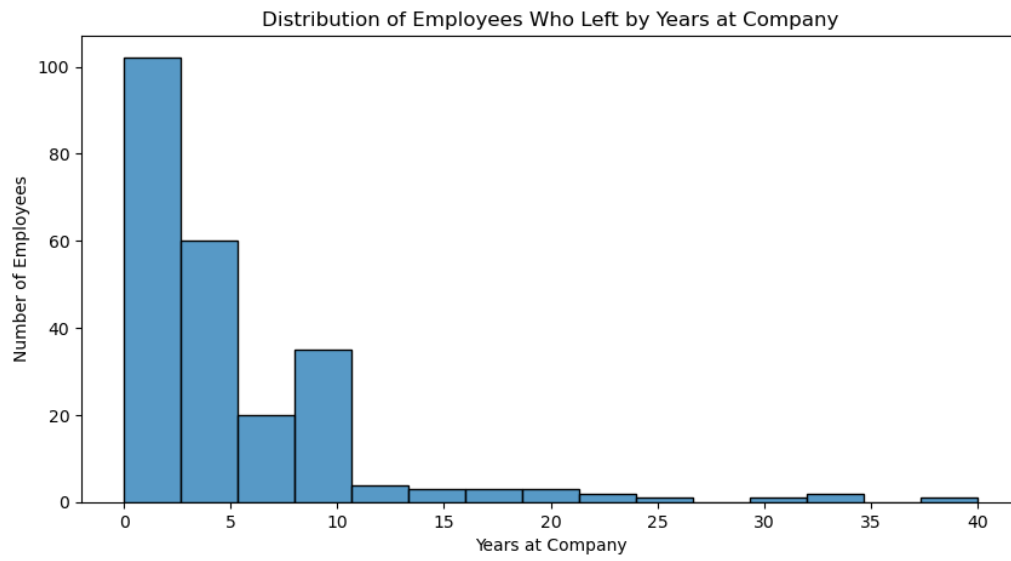
This Machine Learning system predicts whether an employee is likely to leave the company based on factors like job satisfaction, salary, work-life balance, years at the company, and performance ratings.

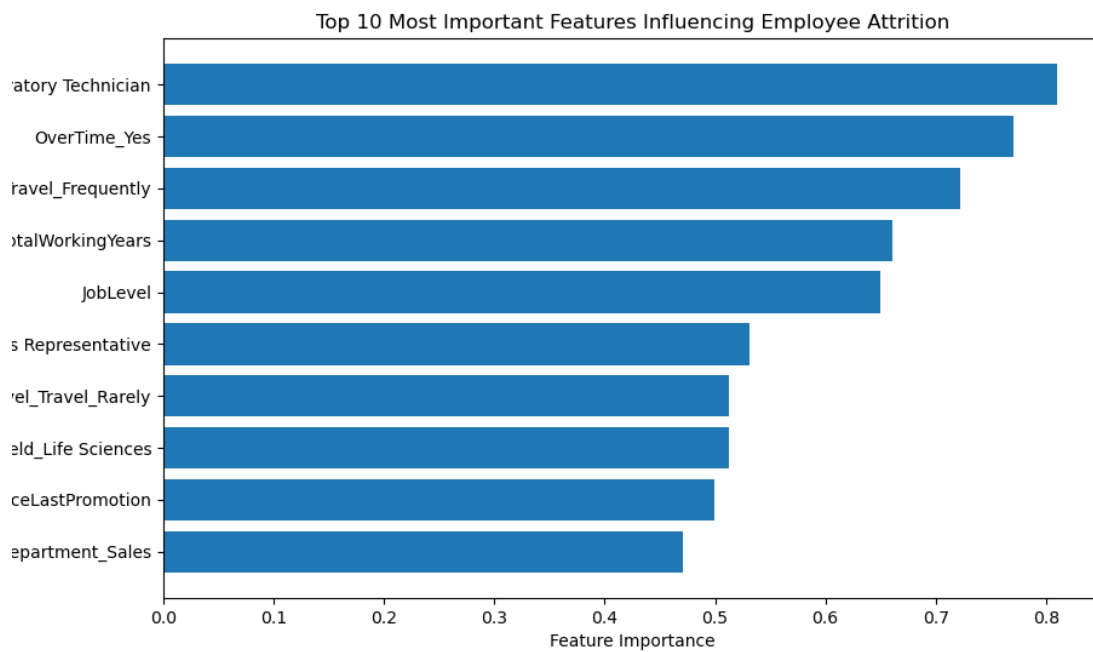
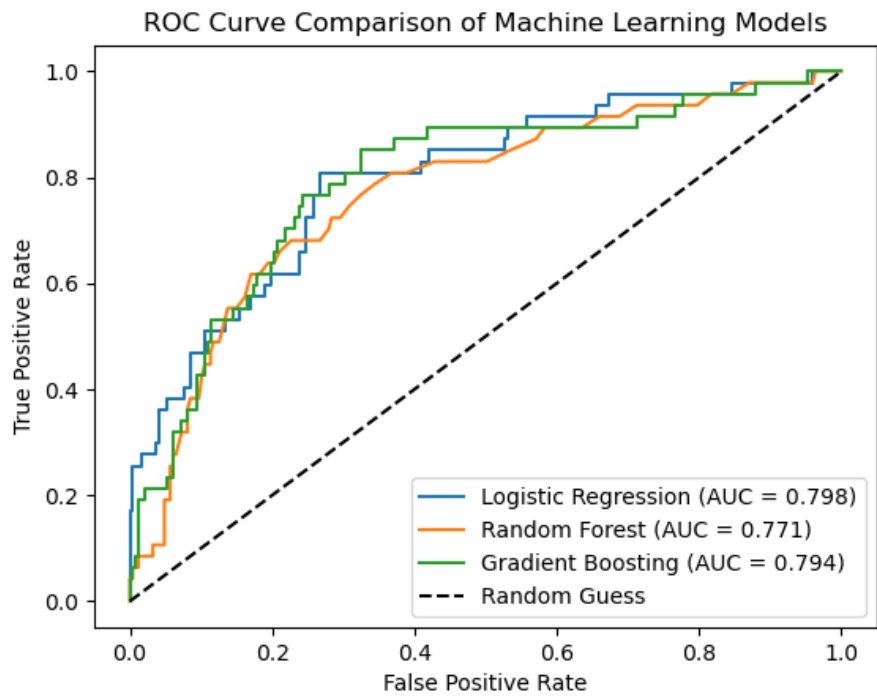
Graphs:











Summary:

Employee attrition is a significant challenge because replacing experienced employees increases recruitment costs, training expenses, and can reduce overall productivity. This project analyzed employee data to identify the key factors associated with employees leaving the organization and to help HR proactively improve retention.

The analysis found that employee turnover is concentrated within specific departments and job roles rather than being evenly distributed across the company. The **Sales department** experienced the highest attrition rate (approximately **20.6%**), while **Sales Representatives** had the highest turnover among all job roles, with nearly **40%** leaving the organization. **Laboratory Technicians** also showed relatively high turnover, indicating that retention efforts should focus on these groups first.

Work-life balance emerged as an important factor. Employees who reported the poorest work-life balance had an attrition rate of approximately **31.2%**, which was substantially higher than employees reporting better work-life balance. Employees who left the company also tended to have **shorter tenures**, with most departures occurring during the first few years of employment. This suggests that the early stages of an employee's career are the most critical for retention.

Although employees who left generally earned lower monthly incomes than those who stayed, salary alone does not fully explain why employees leave. Factors such as overtime, frequent business travel, work-life balance, job role, and career progression appear to have a greater combined influence on employee retention than compensation alone.

Based on these findings, the company should prioritize retention initiatives for employees in the **Sales department**, particularly **Sales Representatives** and **Laboratory Technicians**. HR should also identify employees who frequently work overtime or travel for business and conduct regular retention conversations to address workload and work-life balance concerns. Strengthening onboarding, mentoring, and career development programs during the first few years of employment can further reduce employee turnover.

This analysis provides a practical decision-support tool for HR by identifying employees who may be at a higher risk of leaving. However, the results should complement—not replace—human judgment, as employee decisions can also be influenced by personal circumstances and external factors that are not captured in the available data.